VIDEO GAME CREATION

Lesson 3 - Programming Basics II: Take Your Shot

Semester 1, Week 3

- **Objectives:** Explain the template/instance relationship (one object definition, many independent runtime copies); explain why spawned objects must be explicitly cleaned up; explain the difference between an edge-triggered ("just pressed") and level-triggered ("currently held") input check; add a working shoot mechanic to the ongoing Semester 1 project.
- **IGDA Tie-In:** Programming - object instancing, resource/memory management, input handling (introductory)
- **Featured Engine (lecture demo):** Godot Engine
- **Materials:** Demo laptop with all 8 engines verified (see Teacher's Checklist); Student Handouts, one per engine; students' own Lesson 2 projects (already saved from last week)

Agenda

0:00-0:05	Welcome back, Lesson 2 recap, quick share of any finished Key Press Reactors
0:05-0:25	Instances: One Template, Many Copies (live demo in Godot)
0:25-0:45	Cleanup: Why You Have to Remove What You Create
0:45-0:50	One Press, One Shot (edge-triggered vs. held-down input)
0:50-1:00	Introduce lab exercise, hand out engine-specific handouts
1:00-1:50	LAB: Add shooting to your ongoing [YourName]Game project
1:50-2:00	Exit check / quick share-out

Levels for this lesson's exercise

- Level 1 (Minimal): One key press spawns one projectile that flies upward and disappears off-screen.
- Level 2 (Complete): Firing repeatedly produces independent projectiles, capped by a fire-rate cooldown.
- Level 3 (Refined): Add a fire sound, a distinct projectile look, and/or clean spawn/move/cleanup structure.

See: Lesson 3 Teacher's Script, Lesson 3 Prep Checklist, and the per-engine Student Handouts.